AGA KHAN UNIVERSITY EXAMINATION BOARD
SECONDARY SCHOOL CERTIFICATE CLASS IX EXAMINATION

Chemistry Paper I (Theory & ATP) — Annual 2022

Total Time: 1 hour 40 minutes | Total Marks: 50 (40 Theory & 10 Alternate to Practical)

INSTRUCTIONS: Answer all fifty multiple choice questions. Choose ONE best answer for each question.

Q.1 The formula of an ionic compound containing a metal ion (M+3) and a non-metal ion (X–1) is

A. 3MX.

B. M3X.

C. MX3.

D. M3X3.

Q.2 Which of the following is NOT an empirical formula?

A. CH2

B. CH2O

C. C2H4N2

D. C3H6O2

Q.3 A sample of hydrocarbon undergoes complete combustion to produce CO2 and H2O. If 8.80 g of carbon dioxide is obtained, then what will be the mass of carbon present in it? (C=12, O=16, H=1)

A. 2.4 g

B. 3.8 g

C. 12.0 g

D. 32.3 g

Q.4 The values of w, x, y and z in the chemical equation: w Al(s) + x CuSO4(aq) -> y Cu(s) + z Al2(SO4)3(aq) are:

A. w=2, x=1, y=1, z=2

B. w=2, x=3, y=2, z=1

C. w=2, x=3, y=3, z=1

D. w=3, x=2, y=2, z=1

Q.5 Which type of chemical reaction is shown in the given equation? H2S(g) + Cl2(g) -> S(s) + 2HCl(g)

A. Combustion

B. Decomposition

C. Single displacement

D. Double displacement

Q.6 The reactants of a double displacement reaction are

A. two elements.

B. two ionic compounds.

C. two diatomic molecules.

D. an element and an ionic compound.

Q.7 Which of the following actions will result in a combustion reaction?

A. Freezing a juice bottle

B. Lighting a scented candle

C. Cutting an apple into slices

D. Mixing iron filings with sand

Q.8 Deuterium and tritium are atoms of the same element, i.e. hydrogen. They both have the same:

A. I and II (atomic mass & atomic number)

B. I and III (atomic mass & neutrons)

C. II and IV (atomic number & electrons)

D. III and IV (neutrons & electrons)

Q.9 If an element has 3 electrons in its valence (M) shell, then what will be its atomic number?

A. 15

B. 13

C. 10

D. 5

Q.10 How many electrons, protons and neutrons are there in a nitrogen (14_7 N^–3) ion?

A. Electrons: 4 | Protons: 7 | Neutrons: 10

B. Electrons: 7 | Protons: 7 | Neutrons: 7

C. Electrons: 10 | Protons: 7 | Neutrons: 7

D. Electrons: 17 | Protons: 14 | Neutrons: 7

Q.11 The electronic configuration provides information about all of the following features of an atom EXCEPT the

A. group and period it belongs to.

B. type of chemical bond it can form.

C. presence of neutrons in the nucleus.

D. distribution of electrons in different shells.

Q.12 According to Rutherford atomic model, the nucleus in an atom is

A. large and hard.

B. small and dense.

C. neutral and hollow.

D. light and revolving.

Q.13 In nuclear reactors, the radioactive isotope used to produce energy is

A. cobalt-60.

B. iodine-131.

C. strontium-90.

D. uranium-235.

Q.14 Given are the characteristics of an element X: A salt former, Found in liquid state, Reddish brown in colour, Exists as a diatomic molecule. The element X identified is

A. sodium.

B. chlorine.

C. bromine.

D. beryllium.

Q.15 The electronic configuration of an element of group IIA and period 3 is

A. 1s2, 2s1

B. 1s2, 2s2, 2p1

C. 1s2, 2s2, 2p3

D. 1s2, 2s2, 2p6, 3s2

Q.16 With reference to the different groups of the modern periodic table, the information that stands CORRECT is that all the elements in group

A. 13 are metals.

B. 16 form 2+ cations.

C. 17 produce basic oxides.

D. 18 exist in gaseous state.

Q.17 When an atom M combines with another atom X, an ionic bond is formed. In this reaction, the atomic radius of the atom X increases because

A. it loses electrons to atom M.

B. it gains electrons from atom M.

C. its protons get tightly packed inside the nucleus.

D. its ionisation energy increases on forming the ionic bond.

Q.18 The quantity of energy that an isolated, gaseous atom in the ground electronic state must absorb to discharge an electron is termed as

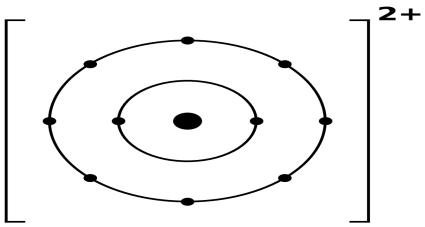
A. kinetic energy.

B. shielding effect.

C. electron affinity.

D. ionisation energy.

Q.19 The given ion is formed from an element belonging to group:



A. IIA.

B. IVA.

C. VIA.

D. VIIIA.

Q.20 The ionic compound whose both ions represent the same electronic configuration as that of neon atom (10_Ne) is

A. K2O

B. KCl

C. MgO

D. MgCl2

Q.21 Which of these elements will gain an electron to form an anion?

A. Boron

B. Argon

C. Lithium

D. Chlorine

Q.22 Species P has 8 protons, 8 neutrons, and electronic structure 2, 8. Species Q has 16 protons, 16 neutrons, and electronic structure 2, 8, 8. Species P and Q are:

A. atoms of inert gases.

B. positively charged ions.

C. negatively charged ions.

D. isotopes of the same element.

Q.23 Which of the following statements stands TRUE for the formation of lithium chloride?

A. A lithium atom gives an electron to the chlorine atom.

B. A chlorine atom gives an electron to the lithium atom.

C. A chlorine atom shares an electron with the lithium atom.

D. A lithium atom accepts a pair of electrons from the chlorine atom.

Q.24 Which set of property and example is TRUE for polar covalent compounds?

A. Electrons are pulled equally | Example: CCl4

B. Electrons are pulled unequally | Example: CCl4

C. Electrons are pulled equally | Example: HCl

D. Electrons are pulled unequally | Example: HCl

Q.25 A coordinate covalent bond is formed when

A. one atom shares an electron pair with another atom.

B. one atom loses electrons and the other atom acquires them.

C. two like atoms share their unpaired electrons with each other.

D. two unlike atoms share their unpaired electrons with each other.

Q.26 Which type of attractive forces are present between I2 molecules?

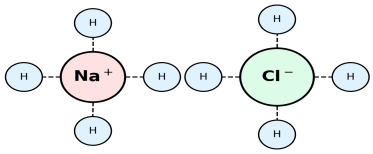
A. Ionic bonding

B. Dispersion forces

C. Hydrogen bonding

D. Dipole-dipole forces

Q.27 The given figure exemplifies the process of:



A. dissolution.

B. neutralisation.

C. crystallisation.

D. decomposition.

Q.28 Bronze is an alloy consisting of 88% copper and 12% tin. Therefore, it is a solution of

A. solid in solid.

B. liquid in solid.

C. solid in liquid.

D. liquid in liquid.

Q.29 A sugar solution is prepared by dissolving 10 g of sugar in 55 g of water. The mass percent of solvent in the given solution is

A. 15.38%.

B. 18.18%.

C. 81.82%.

D. 84.62%.

Q.30 Particles of a mixture: Large but cannot be seen with naked eyes; Big but can pass through filter paper; Quite stable and do not settle down for a long time. This mixture is:

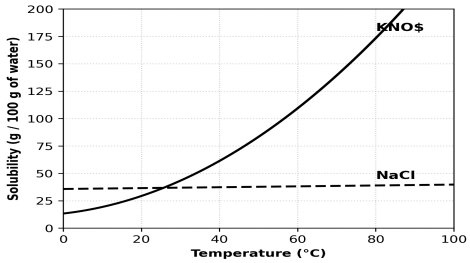
A. lemonade.

B. soap lather.

C. sand in water.

D. milk of magnesia.

Q.31 The given graph shows variation in solubility of KNO3 and NaCl salts with temperature. The TRUE interpretation is:



A. both salts increases with the release of heat.

B. both salts increases with the absorption of heat.

C. NaCl increases while that of KNO3 shows loss of heat.

D. KNO3 increases while that of NaCl shows no change in heat.

Q.32 The method that can be used to investigate the presence of different food colours in a beverage is

A. alloying.

B. crystallisation.

C. decomposition.

D. chromatography.

Q.33 A mixture is formed when chalk is added to water. The particles in this mixture

A. cannot be seen by the naked eye.

B. settle over time if left undisturbed.

C. have a diameter less than 1000 nm.

D. cannot be separated through filtration.

Q.34 The term, 'oxidation' of a substance means

A. gain of electron and loss of oxygen atom.

B. gain of oxygen atom and loss of electron.

C. gain of electron and loss of hydrogen atom.

D. gain of hydrogen atom and loss of electron.

Q.35 The oxidation number of phosphorus in NaH2PO4 is

A. –5

B. –3

C. +3

D. +5

Q.36 Which substance gets reduced when a piece of magnesium is dipped into a copper(II) salt solution?

A. Copper ions

B. Copper atoms

C. Magnesium ions

D. Magnesium atoms

Q.37 Substances: I. Fe, II. $FeCl_2$, III. $FeCl_3$. The substance(s) that can be used as a reducing agent is/ are:

A. I only.

B. I and II.

C. III only.

D. II and III.

Q.38 Solution properties: Does not conduct electricity; Solute is dissolved completely; Solute is not dissociated into ions. The solution is:

A. dilute solution of non-electrolyte.

B. dilute solution of strong electrolyte.

C. concentrated solution of weak electrolyte.

D. concentrated solution of strong electrolyte.

Q.39 During the manufacturing process of caustic soda in Nelson's cell, the by-product that is obtained at anode is/ are

A. chlorine gas.

B. hydrogen gas.

C. sodium metal.

D. water vapours.

Q.40 The term, 'rusting' refers to the corrosion of

A. iron.

B. copper.

C. chromium.

D. aluminium.

Q.41 [ATP] A student used solvent X to separate a mixture of salt and sugar because sugar is soluble in X while salt is not. Solvent X is:

A. oil.

B. water.

C. ethanol.

D. acetic acid.

Q.42 [ATP] In a science laboratory, Mariam prepared a mixture of SiO_2 and alum in water, then filtered it. An inference Mariam must have made is that

A. SiO_2 is insoluble in water.

B. SiO_2 easily passes through filter paper.

C. alum is an amorphous solid.

D. alum easily settles down in water.

Q.43 [ATP] A student accidentally dropped some iron filings in a bottle of sodium chloride. Which step(s) should he/she follow to remove iron filings?

A. Use a magnet only

B. Dissolve mixture in water and filter it

C. Dissolve in water, filter and crystallise filtrate

D. Use a magnet, dissolve in water, filter and evaporate

Q.44 [ATP] When sucrose is dissolved in water, its solubility increases with rise in temperature because sucrose

A. evolves heat in water.

B. ionises completely in water.

C. absorbs heat from the water.

D. reacts chemically with water.

Q.45 [ATP] Oxalic acid is a primary standard substance because of its high

A. density.

B. stability.

C. solubility.

D. reactivity.

Q.46 [ATP] Altamash is provided with a 250 mL volumetric flask to prepare 0.1 M oxalic acid solution ($H_2C_2O_4.2H_2O$). How much mass does he require? (C=12, H=1, O=16)

A. 2.25 g

B. 3.15 g

C. 25.0 g

D. 50.4 g

Q.47 [ATP] In preparing crystals of blue vitriol ($CuSO_4.5H_2O$), the purpose of adding dilute sulphuric acid is to

A. breakdown copper sulphate into its ions.

B. prevent the hydrolysis of copper sulphate.

C. cause decomposition of copper sulphate.

D. remove impurities present in copper sulphate.

Q.48 [ATP] If heating is exceeded (beyond crystallization point) to dryness, this causes:

A. I and II (disappearance of blue colour & loss of water of crystallisation)

B. I and III

C. II and IV

D. III and IV

Q.49 [ATP] Electroplating is a redox reaction to prevent iron from corrosion. All statements are true EXCEPT:

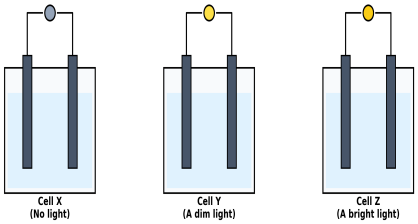
A. transfer of electrons between reactants.

B. use of a salt bridge for the migration of ions.

C. use of electrical energy to initiate reaction.

D. electrolyte of the same metal being coated.

Q.50 [ATP] Electrolytic cells X (no light), Y (dim light), Z (bright light). The electrolytes present in X, Y, Z are:



A. X: potassium chloride, Y: sugar, Z: acetic acid

B. X: sugar, Y: ethanol, Z: hydrochloric acid

C. X: acetic acid, Y: hydrochloric acid, Z: sugar

D. X: ethanol, Y: acetic acid, Z: potassium chloride